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# A Unified Real-Space Frustration Mechanism for Superconductivity: Projective Phase Locking from Conventional to Strongly Correlated Systems

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**Abstract**

We propose a structural interpretation of superconductivity based on projective phase locking of composite configurations within a fiber-constrained geometric framework. Rather than postulating a specific pairing interaction, the mechanism identifies superconductivity with the formation of stable $w = 2$ composites that minimize real-space frustration and enable global phase coherence.

In conventional superconductors, the projective gap is stabilized by mediator-assisted binding, recovering the London and Ginzburg–Landau structure. In strongly correlated materials, pairing emerges from the reduction of staggered frustration, naturally separating amplitude and phase scales and accounting for pseudogap behavior.

The symmetry of the superconducting order parameter is determined by a minimization principle on a fiber raccordement cost functional under lattice and frustration constraints. This criterion predicts $d_{x^2-y^2}$ symmetry in strongly staggered systems such as cuprates, and extended $s^{\pm}$ symmetry in nickelates, where antiferromagnetic frustration is reduced and partially isotropized.

To leading order, the critical temperature scales as $T_c \sim \frac{\pi}{2}\delta J f(r_{\mathcal{F}})$, linking phase stiffness to independently measurable exchange and frustration amplitudes. The framework is explicitly falsifiable through gap symmetry determination, phase stiffness scaling, disorder response, and pressure dependence.

These results suggest that superconductivity may be understood as a geometric consequence of frustration reduction and global fiber coherence, providing a unified structural description of low- and high-$T_c$ regimes.

## 1 Introduction

Superconductivity remains one of the most striking collective phenomena in condensed matter physics [1, 2]. Despite more than a century of experimental and theoretical advances, a unified structural understanding across conventional and strongly correlated regimes remains elusive. In conventional low-$T_c$ materials, Bardeen–Cooper–Schrieffer (BCS) theory successfully explains pairing as a phonon-mediated instability of the Fermi surface [3]. In contrast, high-$T_c$ superconductors such as cuprates and related materials exhibit strong electronic correlations, pseudogap behavior, and unconventional gap symmetries [4, 5] that resist a purely mediator-based description.

The conceptual separation between these regimes has led to a proliferation of mechanisms tailored to specific material families. Phonon exchange, spin fluctuations, orbital fluctuations, and various strong-coupling approaches have been proposed as primary drivers of pairing [6]. While many of these models capture important aspects of experimental data, they typically operate within a momentum-space interaction framework and often rely on material-specific assumptions.

In this work we pursue a different route. We propose that superconductivity can be understood as *projective phase locking* arising from a real-space geometric constraint. Within the Cosmochrony framework [7], electromagnetism is identified with the $U(1)$ fiber of a projective structure $\Pi \cong S^3$ [8], whose phase coordinate $\theta$ encodes the effective gauge degree of freedom. Topologically non-trivial excitations of this fiber correspond to charged solitonic configurations [9]. A bound composite of winding number $w = 2$ emerges as a structurally distinct class, carrying effective charge $2e$ and supporting a collective phase degree of freedom.

The central claim of this paper is that superconductivity results from the stabilization and phase locking of such $w = 2$ composites. In conventional materials, a bosonic mediator such as a phonon lowers the projective gap $\Delta_\Pi$ and stabilizes the composite configuration, consistent with the BCS paradigm [3]. In strongly correlated systems, pairing instead arises as a reduction of *real-space frustration* in the admissibility constraints of the fiber. In both regimes, the London [2] and Ginzburg–Landau [10] structures follow from the cost of breaking global fiber coherence, rather than from an assumed symmetry-breaking potential.

This approach yields a unified structural picture with several consequences. First, it naturally separates amplitude and phase scales: the projective gap governing local composite stability can differ from the global phase-locking temperature, providing a geometric account of pseudogap behavior [4]. Second, the symmetry of the superconducting gap is determined by a real-space minimization problem on the fiber raccordement cost under lattice constraints, rather than by a specific momentum-dependent interaction kernel. Third, the phase stiffness scales with the density of active composites, leading to $\rho_s \propto \delta J$ in strongly correlated quasi-two-dimensional systems, consistent with empirical Uemura scaling [11] and Berezinskii–Kosterlitz–Thouless phase-locking behavior [12], where $\delta$ is the carrier concentration and $J$ the dominant frustration scale.

A central predictive application of this framework concerns superconducting nickelates [13]. Because their antiferromagnetic frustration sector is reduced and partially isotropized relative to cuprates, the fiber-raccordement criterion predicts an $s^\pm$ gap symmetry rather than $d_{x^2-y^2}$ symmetry at optimal doping, for which strong experimental evidence exists in cuprates [5]. The predicted $T_c$ ratio between nickelates and cuprates follows from independently measurable parameters, including magnetic exchange scales and structure factors, without introducing free fitting constants. The predicted intermediate sensitivity to non-magnetic disorder provides a directly testable signature [14].

The purpose of this paper is therefore threefold. We first derive the universal London and Ginzburg–Landau structure from the projective fiber geometry. We then extend the mechanism to the strongly correlated regime through a real-space frustration criterion. Finally, we formulate explicit falsifiable predictions, most notably for nickelate superconductors, and delineate the experimental conditions under which the mechanism would be ruled out.

The framework presented here does not attempt to replace microscopic electronic models. Rather, it proposes a geometric organizing principle that constrains the admissible pairing channels and the scaling of phase coherence. Its validity can therefore be assessed by the consistency of its structural predictions with symmetry selection, stiffness scaling, disorder response, and pressure dependence across material families.

## 2 Structural Framework

### *2.1 Projection Fiber and Effective $U(1)$ Structure*

The present mechanism is formulated within a minimal structural setting. We assume that the effective electromagnetic gauge symmetry arises as the $U(1)$ fiber symmetry of a projective structure

$$\Pi \cong S^3, \tag{1}$$

whose Hopf fibration defines a base manifold and a circular fiber. The fiber coordinate

$$\theta \in [0, 2\pi) \tag{2}$$

plays the role of an effective phase variable.

The electromagnetic potential $A$ is identified with the connection associated with this $U(1)$ fiber in the projected description. It is not an additional gauge field introduced independently, but the connection of the fiber as it appears in the effective theory. Gauge covariance therefore fixes the only admissible covariant gradient for the fiber phase,

$$D\theta \equiv \nabla\theta - \frac{q}{\hbar}A, \tag{3}$$

where $q$ denotes the effective topological charge of the configuration under consideration.

No further gauge postulate is introduced. The $U(1)$ structure is entirely inherited from the fiber geometry.

### *2.2 Topological Excitations and Composite Classes*

Charged excitations correspond to topologically non-trivial configurations of the fiber phase. A single elementary excitation carries winding number

$$w = 1, \tag{4}$$

and couples to the connection with effective charge $q = e$.

A bound composite of two such excitations forms a distinct topological class characterized by

$$w = 2. \tag{5}$$

This class is not simply the superposition of two independent windings. It corresponds to a globally coherent configuration in which the fiber phase interpolates continuously between the two cores, subject to a raccordement constraint. Separating the cores introduces a phase defect whose energetic cost defines a stability threshold.

For the $w = 2$ class, the covariant gradient becomes

$$D\theta = \nabla\theta - \frac{2e}{\hbar}A. \tag{6}$$

This composite degree of freedom therefore carries effective charge $2e$ and supports a collective phase variable $\theta$.

### *2.3 Projective Gap and Stability Criterion*

We define the *projective gap* $\Delta_\Pi$ as the minimal excitation energy required to destabilize the $w = 2$ composite class. Operationally, this corresponds to the lowest positive eigenvalue of a linearized relaxation operator restricted to the $U(1)$ fiber sector around the composite configuration,

$$\Delta_\Pi \equiv \lambda_{\min}\left(\mathcal{L}_\Pi\big|_{w=2}\right). \tag{7}$$

Physically, $\Delta_\Pi$ is the minimal energy cost to either separate the composite into two $w = 1$ excitations or introduce a local phase slip that alters the global winding class. The existence of a finite $\Delta_\Pi$ is the structural condition for composite stability.

Two distinct mechanisms may generate such stability:

- In conventional low-$T_c$ systems, a bosonic mediator lowers the effective raccordement cost, stabilizing the composite through an attractive channel.
- In strongly correlated systems, stability arises from the reduction of real-space frustration in the admissibility constraints of the fiber, without invoking a separate mediator.

The remainder of this work shows that once a stable $w = 2$ composite exists, global phase locking of the fiber naturally generates the London and Ginzburg–Landau structures, flux quantization, and Meissner screening.

### *2.4 Free-Energy Functional from Phase Coherence*

When a macroscopic density of $w = 2$ composites occupies the same coherent phase sector, we define the order parameter

$$\Psi = |\Psi|e^{i\theta}, \tag{8}$$

where $|\Psi|^2$ measures the density of composites participating in coherent projection.

The leading contribution to the free energy must be quadratic in the only gauge-covariant derivative available,

$$\mathcal{F} \supset \frac{\hbar^2\rho_s}{2m^*}\left(\nabla\theta - \frac{2e}{\hbar}A\right)^2, \tag{9}$$

where $\rho_s$ is the phase stiffness and $m^*$ the effective inertia of the composite mode.

Equation 9 is structurally identical to the London and Ginzburg–Landau gradient terms. It arises here from the energetic cost of breaking global fiber coherence, not from an assumed symmetry-breaking potential.

The Meissner effect, flux quantization, and the effective mass of the gauge field follow directly from Eq. 9. These consequences will be developed explicitly in the following sections.

## 3 Low-$T_c$ Regime: Mediator-Stabilized Composites

### *3.1 Mediator-Induced Reduction of the Projective Gap*

In conventional superconductors, pair formation is well described phenomenologically by an effective attractive interaction mediated by lattice vibrations. Within the present framework, this mechanism is reinterpreted as a reduction of the projective gap $\Delta_\Pi$ associated with the $w = 2$ composite class.

The phonon field does not introduce a fundamentally new pairing degree of freedom. Rather, it modifies the local raccordement cost of the fiber phase between neighboring electronic excitations. The effective interaction lowers the minimal eigenvalue of the relaxation operator in Eq. 7, thereby stabilizing the composite configuration.

We denote the renormalized gap as

$$\Delta_\Pi^{\mathrm{eff}} = \Delta_\Pi^{(0)} - \delta\Delta_\Pi^{(\mathrm{ph})}, \tag{10}$$

where $\Delta_\Pi^{(0)}$ is the intrinsic stability cost in the absence of mediator coupling, and $\delta\Delta_\Pi^{(\mathrm{ph})}$ encodes the phonon-induced reduction.

A finite composite population becomes thermodynamically stable when

$$k_B T \lesssim \Delta_\Pi^{\mathrm{eff}}. \tag{11}$$

Equation 11 defines the onset of local composite formation, which coincides with the superconducting transition in weakly correlated materials.

### *3.2 Emergence of the BCS Gap Structure*

Although the structural origin of pairing differs from the standard interaction-based description, the resulting excitation spectrum is equivalent.

In the coherent regime, breaking a composite requires at least energy $2\Delta_\Pi^{\mathrm{eff}}$, corresponding to the creation of two $w = 1$ excitations. The quasiparticle dispersion therefore takes the form

$$E(\mathbf{k}) = \sqrt{\xi_{\mathbf{k}}^2 + \Delta^2}, \tag{12}$$

with

$$\Delta \equiv \Delta_\Pi^{\mathrm{eff}}. \tag{13}$$

The formal mapping between the projective gap $\Delta_\Pi$ and the quasiparticle gap $\Delta(\mathbf{k})$ measured by ARPES is derived in Appendix C.

The conventional $s$-wave symmetry corresponds to the isotropic $w = 2$ composite class that minimizes the raccordement cost in a lattice without strong anisotropic frustration. No additional angular structure is required, and the gap remains nodeless across the Fermi surface.

The exponential dependence of $\Delta$ on the effective coupling strength in BCS theory is reinterpreted here as the sensitivity of the minimal eigenvalue $\lambda_{\min}(\mathcal{L}_\Pi)$ to weak mediator-induced renormalization of the fiber stability cost.

### *3.3 London Electrodynamics and Meissner Screening*

Once a macroscopic fraction of composites occupies a common phase sector, the free-energy functional of Eq. 9 governs long-wavelength electrodynamics.

Varying the free energy with respect to $A$ yields the London equation

$$\nabla^2 \mathbf{B} = \frac{1}{\lambda_L^2}\mathbf{B}, \tag{14}$$

with penetration depth

$$\lambda_L^{-2} = \frac{4e^2\rho_s}{\hbar^2 m^*}. \tag{15}$$

The magnetic field is therefore expelled from the bulk over the scale $\lambda_L$, demonstrating the Meissner effect. In this interpretation, the gauge field acquires an effective mass through phase locking of the fiber, analogous to the Anderson–Higgs mechanism in condensed matter systems.

### *3.4 Flux Quantization*

A vortex corresponds to a topological defect in the fiber phase. Around a closed contour encircling the vortex core,

$$\oint \nabla\theta \cdot d\ell = 2\pi n, \qquad n \in \mathbb{Z}. \tag{16}$$

Imposing $D\theta = 0$ outside the core, one obtains

$$\oint A \cdot d\ell = \frac{\hbar}{2e} \oint \nabla\theta \cdot d\ell = n\frac{h}{2e}. \tag{17}$$

The magnetic flux is therefore quantized in units

$$\Phi_0 = \frac{h}{2e}. \tag{18}$$

Flux quantization follows directly from the topological winding of the fiber phase and does not require an independent postulate.

### *3.5 Summary of the Low-$T_c$ Regime*

In the conventional regime, a mediator lowers the projective gap and stabilizes an isotropic $w = 2$ composite. Global phase locking of these composites generates the London and Ginzburg–Landau structures, the Meissner effect, and flux quantization.

The standard BCS phenomenology is therefore recovered as a special case of mediator-stabilized projective phase locking. The essential structural ingredient is not the specific form of the interaction, but the existence of a stable $w = 2$ composite class and the energetic cost of breaking global fiber coherence.

## 4 Strongly Correlated Regime: Frustration-Stabilized Composites

### *4.1 Real-Space Frustration and Raccordement Cost*

In strongly correlated materials such as cuprates, the normal state cannot be described as a weakly interacting Fermi liquid. At low doping, the system is proximate to a Mott insulating phase with robust antiferromagnetic correlations. Within the present framework, this regime is characterized by a high density of real-space admissibility constraints on the fiber.

We denote by $\mathcal{F}$ the local frustration density, defined operationally as a proxy for the fraction of nearest-neighbor raccordement constraints that cannot be simultaneously satisfied. A quantitative microscopic definition and its experimental grounding are provided in Appendix A. In cuprates, this quantity is tied to the antiferromagnetic structure factor $S(\pi, \pi)$, measurable by neutron scattering or RIXS, and we take

$$\mathcal{F} \propto \frac{S(\pi, \pi)}{S_{\max}}, \tag{19}$$

where $S_{\max}$ denotes the fully ordered reference value.

In this regime, isolated $w = 1$ excitations carry a high raccordement cost, as their presence locally increases the incompatibility of neighboring fiber phases. A bound $w = 2$ composite, however, can reduce the global frustration by redistributing the phase mismatch over a larger region. Pair formation is therefore not driven by an external mediator, but by the minimization of the real-space fiber incompatibility.

The projective gap $\Delta_{\Pi}^{\text{amplitude}}$ in this regime is set by the dominant frustration scale, which in cuprates is the magnetic exchange energy $J$. To leading order,

$$\Delta_{\Pi}^{\text{amplitude}} \sim J. \tag{20}$$

### *4.2 Symmetry Selection from Lattice Constraints*

The symmetry of the superconducting gap is determined by minimizing the total raccordement cost of the composite configuration under lattice symmetry constraints.

In cuprates, the underlying lattice symmetry is $D_{4h}$, and the dominant frustration vector is $(\pi, \pi)$. The staggered nature of this frustration imposes a sign-alternating constraint on the fiber phase between neighboring plaquettes.

The admissible composite must satisfy two conditions:

1. minimize overlap with the frustrated $(\pi, \pi)$ sector,
2. preserve global fiber continuity without introducing defects.

Among the irreducible representations of $D_{4h}$ available to a $w = 2$ composite, the lowest-cost solution under these constraints lies in the $B_{1g}$ sector. Its nodal structure aligns with the $(\pi, 0)$ and $(0, \pi)$ directions, distributing the required sign alternation across the Fermi surface while avoiding localized raccordement singularities. The explicit mapping between the projective spectrum $\Delta_\Pi(\mathbf{k})$ and the ARPES gap structure underlying this nodal behavior is derived in Appendix C.

Extended $s$-wave configurations do not simultaneously satisfy both conditions. Although certain $A_{1g}$ form factors change sign, they either accumulate excessive cost near the $(\pi, \pi)$ sector or require additional fiber defects to maintain global continuity.

The selection of $d_{x^2-y^2}$ symmetry thus follows from a real-space geometric minimization problem, rather than from a momentum-space interaction kernel.

### *4.3 Amplitude–Phase Separation and the Pseudogap*

A crucial distinction between conventional and strongly correlated regimes is the separation of amplitude and phase scales.

The formation of local $w = 2$ composites occurs when

$$k_B T \lesssim \Delta_\Pi^{\text{amplitude}} \sim J. \tag{21}$$

We identify this scale with the pseudogap temperature $T^*$.

However, global superconductivity requires phase locking across the system. The relevant quantity is the phase stiffness $\rho_s$, which measures the energetic cost of spatial variation of the fiber phase.

In the doped regime, the density of active composites is proportional to the carrier concentration $\delta$. The stiffness therefore scales as

$$\rho_s^{(0)} \propto \delta\, \Delta_\Pi^{\text{amplitude}} \sim \delta J. \tag{22}$$

The superconducting transition temperature is governed by phase coherence rather than amplitude formation. In quasi-two-dimensional systems, the transition is controlled by vortex unbinding, leading to the Berezinskii–Kosterlitz–Thouless criterion

$$k_B T_c \simeq \frac{\pi}{2} \rho_s(T_c^-) \sim \frac{\pi}{2} \delta J, \tag{23}$$

up to corrections from interlayer coupling.

This naturally explains the separation between $T^*$ and $T_c$, as well as the linear scaling of $\rho_s$ with doping observed experimentally (Uemura relation).

### *4.4 Dome Structure and Frustration Ratio*

The superconducting dome arises from the competition between frustration amplitude and phase stiffness.

At very low doping, $\mathcal{F}$ is large, but $\delta$ is small, yielding strong amplitude formation yet insufficient phase stiffness. The system remains antiferromagnetic.

At optimal doping, $\mathcal{F}$ is reduced while $\delta$ is sufficient to sustain large $\rho_s$, maximizing $T_c$.

In the overdoped regime, $\mathcal{F}$ becomes small, reducing the intrinsic stabilization of the composite, and $T_c$ decreases accordingly.

The ratio

$$r_\mathcal{F} \equiv \text{first-order proxy for relative frustration amplitude} \tag{24}$$

will play a central role in comparing material families, as developed in Section 7.

## 5 Phase Stiffness, Quasi-Two-Dimensional Coherence, and Operational Proxies

### *5.1 Phase Stiffness as Relaxation Capacity*

In the strongly correlated regime, the decisive quantity controlling the superconducting transition is not the amplitude gap $\Delta_\Pi^{\text{amplitude}}$, but the phase stiffness $\rho_s$.

Within the projective framework, $\rho_s$ measures the relaxation capacity of the projected substrate, that is, its ability to redistribute local fiber phase mismatches without generating topological defects. We therefore identify $\rho_s$ as the operational proxy for the relaxation flux

$$\mathcal{R} \equiv \rho_s. \tag{25}$$

In doped Mott systems, the density of active composites is proportional to the carrier concentration $\delta$. Combined with Eq. 20, this yields to leading order

$$\rho_s^{(0)} \propto \delta J. \tag{26}$$

Equation 26 provides the structural origin of the experimentally observed Uemura scaling in underdoped cuprates.

### *5.2 Quasi-Two-Dimensional Phase Transition*

Cuprates and related materials are strongly anisotropic layered systems. The dominant projective substrate is provided by the $CuO_2$ planes, while the interlayer coupling $J_\perp$ is significantly smaller than the in-plane exchange scale $J$.

In this quasi-two-dimensional limit, phase coherence is established within each layer through vortex unbinding. The superconducting transition is therefore governed by the Berezinskii–Kosterlitz–Thouless (BKT) criterion,

$$k_B T_c \simeq \frac{\pi}{2}\rho_s(T_c^-), \tag{27}$$

where $\rho_s(T_c^-)$ denotes the renormalized stiffness evaluated at the vortex scale.

The system is not strictly two-dimensional. Finite interlayer coupling converts the BKT transition into a true three-dimensional transition at a slightly higher temperature. The quasi-2D description applies in the strongly anisotropic regime

$$J_\perp \ll k_B T_c \ll J, \tag{28}$$

which is satisfied in cuprate families over a broad doping range.

### *5.3 Frustration Density as Observable Proxy*

To avoid purely definitional constructs, the frustration density $\mathcal{F}$ must be independently accessible from experiment.

We use as first-order proxy

$$\mathcal{F} \propto \frac{S(\pi,\pi)}{S_{\max}}, \tag{29}$$

where $S(\pi,\pi)$ is the magnetic structure factor measured by neutron scattering or RIXS, and $S_{\max}$ is the fully ordered reference value.

This proportionality is a leading-order assumption. Non-linear corrections would modify numerical prefactors but do not alter the qualitative phase selection mechanism unless fine-tuned cancellations occur.

### *5.4 Three Regimes from the Ratio $\mathcal{F}/\mathcal{R}$*

The interplay between frustration density $\mathcal{F}$ and relaxation capacity $\mathcal{R} = \rho_s$ defines three qualitatively distinct regimes.

1. **Antiferromagnetic regime:** $\mathcal{F} \to 1, \quad \rho_s \to 0$. Long-range AF order dominates, and composites cannot delocalize coherently.
2. **Superconducting regime:** Intermediate $\mathcal{F}$ with finite $\rho_s$. Frustration is sufficient to stabilize composites, while relaxation capacity allows global phase locking.
3. **Glass-like or incoherent metallic regime:** $\mathcal{F} \to 0, \quad \rho_s \to 0$ with strong disorder. Frustration is too weak or too fragmented to stabilize composites coherently.

These regimes are experimentally distinguishable. A material exhibiting intermediate $S(\pi,\pi)$ but vanishing $\rho_s$ must possess an additional mechanism suppressing phase coherence, such as strong disorder or orbital mixing. Conversely, superconductivity in a system with negligible $(\pi,\pi)$ frustration requires a different frustration channel, providing a falsifiable diagnostic criterion.

### *5.5 Emergent Dome Structure*

The superconducting dome arises naturally from the doping dependence of both $\mathcal{F}$ and $\rho_s$.

At low doping, $\mathcal{F}$ is large but $\rho_s$ is small. At optimal doping, $\mathcal{F}$ is reduced while $\rho_s \propto \delta J$ is maximal. At overdoping, $\mathcal{F}$ becomes too small to sustain a significant projective gap, and $T_c$ decreases accordingly.

No additional phenomenological tuning parameter is required to generate the dome structure. It emerges from the coupled evolution of frustration amplitude and phase stiffness.

## 6 Symmetry Selection Across Material Families

### *6.1 General Symmetry Selection Principle*

Within the projective framework, the superconducting gap symmetry is not postulated. It emerges from a minimization problem on the fiber raccordement cost under lattice and frustration constraints.

Given a lattice symmetry group $G$ and a dominant frustration channel characterized by a wavevector $\mathbf{Q}$, the admissible $w = 2$ composite configurations belong to irreducible representations of $G$. The selected representation is the one that minimizes the total fiber raccordement cost functional,

$$\mathcal{C}_{\mathrm{tot}} = \int_{\mathrm{FS}} \mathcal{C}_{\Pi}\big(\mathbf{k}; \mathbf{Q}, \mathcal{F}\big)\, d\Omega, \tag{30}$$

subject to global fiber continuity. The explicit variational derivation of this minimization is given in Appendix B.

The functional $\mathcal{C}_{\Pi}$ encodes the local mismatch between the composite phase configuration and the staggered frustration background. It depends on the dominant frustration vector $\mathbf{Q}$ and on the frustration amplitude proxy $\mathcal{F}$.

The symmetry channel minimizing Eq. 30 defines the superconducting gap representation.

### *6.2 Role of the Frustration Ratio*

To compare material families, we introduce the dimensionless frustration ratio

$$r_{\mathcal{F}} \equiv \text{first-order proxy for relative frustration amplitude}, \tag{31}$$

taken to scale proportionally, at leading order, with the magnetic structure factor $S(\mathbf{Q})$.

The ratio $r_{\mathcal{F}}$ does not fix the symmetry by itself. It controls the relative weight of the staggered constraint in the cost functional $\mathcal{C}_{\Pi}$.

Two limiting regimes can be identified:

- **Strongly staggered regime** ($r_{\mathcal{F}}$ large): the composite must strongly suppress overlap with the dominant frustration vector. Nodal structures aligned with $\mathbf{Q}$ become energetically favorable.
- **Weak or isotropized frustration regime** ($r_{\mathcal{F}}$ small): the staggered constraint is reduced, and isotropic or extended $s$-wave configurations minimize the global cost.

In both cases, the lattice symmetry group $G$ restricts the admissible irreducible representations. The interplay between $G$ and $r_{\mathcal{F}}$ determines the selected gap symmetry.

### *6.3 Material Comparison Criterion*

Consider two material families with comparable lattice symmetry but different frustration amplitudes. Let their dominant exchange scales be $J_1$ and $J_2$, and their frustration ratios $r_{\mathcal{F}}^{(1)}$ and $r_{\mathcal{F}}^{(2)}$.

To leading order, the amplitude scale of composite formation is set by $J$, while the symmetry channel is determined by the minimization of $\mathcal{C}_{\Pi}$ under the corresponding $r_{\mathcal{F}}$.

The transition temperature then scales as

$$T_c \sim \frac{\pi}{2}\delta J \cdot f(r_{\mathcal{F}}), \tag{32}$$

where $f(r_{\mathcal{F}})$ is a monotonic function encoding the effect of frustration amplitude on phase stiffness and composite stability.

The precise functional form of $f$ is model-dependent. However, its qualitative behavior is constrained:

- $f(r_{\mathcal{F}})$ increases from zero at vanishing frustration to a maximum at intermediate $r_{\mathcal{F}}$.
- For very large $r_{\mathcal{F}}$, phase stiffness is suppressed by reduced mobility, and $T_c$ decreases.

Equation 32 provides a structural comparison tool between related compounds.

*6.4 Preparation for Nickelates*

Cuprates correspond to a regime of strong $(\pi, \pi)$ frustration and large $J$, yielding $d_{x^2-y^2}$ symmetry as the lowest-cost representation.

Nickelates share similar lattice symmetry but exhibit reduced antiferromagnetic correlations, modified Fermi surface topology, and a smaller exchange scale.

Within the present framework, these differences imply:

1. a reduced $r_{\mathcal{F}}$ relative to cuprates,
2. a modification of the minimization landscape of $\mathcal{C}_{\Pi}$,
3. a shift in the preferred irreducible representation.

The next section develops the resulting quantitative and qualitative predictions.

## 7 Application: Nickelate Superconductors

*7.1 Structural Differences Relative to Cuprates*

Infinite-layer nickelates $R_{1-x}\mathrm{Sr}_x\mathrm{NiO}_2$ share the same square lattice symmetry ($D_{4h}$) as cuprates, but differ in several structural respects.

First, the antiferromagnetic structure factor at wavevector $(\pi, \pi)$ is significantly reduced relative to cuprates. RIXS and neutron measurements indicate a smaller magnetic exchange scale $J_{\mathrm{Ni}}$ and a weaker peak intensity $S_{\mathrm{Ni}}(\pi, \pi)$.

Second, the Fermi surface is multi-band. In addition to the Ni-derived band, an electron pocket near $\Gamma = (0, 0)$ originating from rare-earth orbitals modifies the nesting properties and partially isotropizes the electronic structure.

Within the framework of Section 6, these differences imply a reduced frustration ratio

$$r_{\mathcal{F}}^{\mathrm{Ni}} < r_{\mathcal{F}}^{\mathrm{Cu}}, \tag{33}$$

where $r_{\mathcal{F}}$ is taken as a first-order proxy proportional to $S(\pi, \pi)$. A quantitative definition of this proxy and representative experimental values are given in Appendix A.

*7.2 Predicted Gap Symmetry*

The symmetry channel is determined by minimizing the cost functional $\mathcal{C}_{\Pi}$ under the reduced and partially isotropized frustration constraint.

In cuprates, the strong staggered $(\pi, \pi)$ frustration selects the $B_{1g}$ representation ($d_{x^2-y^2}$ symmetry). In nickelates, the reduced frustration amplitude and the presence of a $\Gamma$ pocket alter the minimization landscape.

The framework predicts that the lowest-cost configuration at optimal doping lies in an extended $s$-wave class, specifically an $s^{\pm}$ structure with the following characteristics:

1. The gap is nodeless within each Fermi pocket at optimal doping.
2. A sign change occurs between the Ni-derived and rare-earth-derived pockets.
3. In the underdoped regime, nodal structure may develop near pocket edges as $r_{\mathcal{F}}$ increases and the staggered constraint regains angular weight.

Many spin-fluctuation approaches yield either $d$-wave or multi-band $s^{\pm}$ depending on assumed Fermiology and interaction strength, with no current consensus. The present mechanism differs in that symmetry selection is controlled by the real-space frustration ratio rather than by a momentum-dependent interaction kernel.

*7.3 Quantitative Estimate of the Critical Temperature*

To leading order, the transition temperature in the quasi-2D regime obeys

$$k_B T_c \simeq \frac{\pi}{2} \delta J \, f(r_{\mathcal{F}}), \tag{34}$$

as introduced in Eq. 32.

For nickelates, using representative experimental values

$$J_{\mathrm{Ni}} \approx 60\text{–}80\,\mathrm{meV}, \qquad \delta_{\mathrm{opt}}^{\mathrm{Ni}} \approx 0.20, \tag{35}$$

and taking $r_{\mathcal{F}}^{\rm Ni}$ as a first-order proxy for the reduced $(\pi,\pi)$ structure factor, one obtains

$$\frac{T_c^{\rm Ni}}{T_c^{\rm Cu}} \approx \frac{\delta_{\rm Ni} J_{\rm Ni}}{\delta_{\rm Cu} J_{\rm Cu}} \cdot \frac{r_{\mathcal{F}}^{\rm Ni}}{r_{\mathcal{F}}^{\rm Cu}}, \tag{36}$$

up to order-unity corrections.

Using representative cuprate values $J_{\rm Cu} \approx 120$–$150\,\mathrm{meV}$ and $\delta_{\rm opt}^{\rm Cu} \approx 0.16$, this estimate yields

$$T_c^{\rm Ni} \sim 0.3\text{–}0.4\, T_c^{\rm Cu}, \tag{37}$$

consistent with observed nickelate transition temperatures in the 10–40 K range, and potentially higher under pressure.

This estimate involves no free fitting parameter. All quantities entering the ratio ($J$, $\delta$, $S(\pi,\pi)$) are independently measurable.

### *7.4 Disorder and Pressure Signatures*

The predicted $s^{\pm}$ symmetry implies intermediate sensitivity to non-magnetic disorder.

Specifically, the suppression rate

$$\frac{dT_c}{dn_{\rm imp}} \tag{38}$$

under non-magnetic impurity substitution should lie between the Abrikosov–Gor'kov $d$-wave limit and the Anderson-theorem $s^{++}$ limit. Systematic impurity studies therefore provide a direct test of the mechanism.

Under pressure, both the effective exchange scale $J_{\rm Ni}$ and the inter-orbital hybridization are modified, entering the framework through $J$ and $r_{\mathcal{F}}$. An enhancement of $T_c$ correlated with measurable changes in magnetic excitation spectra would support the structural interpretation.

### *7.5 Experimental Falsifiability*

The mechanism would be invalidated if:

- A robust $d_{x^2-y^2}$ gap symmetry is conclusively established in nickelates despite reduced $(\pi,\pi)$ frustration.
- The measured suppression of $T_c$ under non-magnetic disorder matches precisely either pure $d$-wave or fully Anderson-protected $s^{++}$ behavior.
- $T_c$ enhancement under pressure occurs without measurable modification of magnetic excitation scales or frustration proxies.

The predictive content of the framework lies in the enforced coupling between frustration amplitude, symmetry selection, and phase stiffness scaling.

## 8 Discussion and Outlook

### *8.1 Scope of the Mechanism*

The framework developed in this work proposes a structural interpretation of superconductivity as projective phase locking of $w = 2$ composite configurations. It does not replace microscopic electronic Hamiltonians. Rather, it provides a geometric organizing principle that constrains admissible pairing channels, phase stiffness scaling, and symmetry selection across material families.

In conventional superconductors, the mechanism reduces to mediator-stabilized composite formation, recovering the standard London and Ginzburg–Landau structure. In strongly correlated systems, pairing arises from the reduction of real-space frustration in fiber admissibility constraints, naturally separating amplitude and phase scales.

The resulting picture unifies low-$T_c$ and high-$T_c$ regimes without introducing distinct pairing principles. Only the origin of the projective gap $\Delta_\Pi$ differs between material classes.

### *8.2 Relation to Existing Theoretical Approaches*

The present framework differs from momentum-space interaction models in its real-space formulation. In particular, symmetry selection is determined by minimization of a fiber raccordement cost under lattice and frustration constraints, rather than by the structure of an interaction kernel $V(\mathbf{k},\mathbf{k}')$.

This distinction does not invalidate spin-fluctuation or strong-coupling approaches. Instead, it constrains their admissible outcomes. If the experimentally observed gap symmetry is inconsistent with the frustration ratio in a given material, the geometric criterion proposed here would be falsified.

The framework therefore operates at a structural level above specific microscopic models. Its role is to provide a symmetry and scaling constraint, not a detailed electronic mechanism.

### *8.3 Limitations*

Several limitations must be acknowledged.

First, the cost functional $\mathcal{C}_\Pi$ is not derived from a fully microscopic $\chi$-dynamics in the present work. Only its symmetry properties and leading scaling behavior are specified. A microscopic derivation remains an open program.

Second, the proportionality between frustration density and magnetic structure factor is introduced as a first-order proxy. Although this assumption is physically motivated, non-linear corrections may alter numerical prefactors. The robustness of predictions therefore rests on qualitative rather than fine-tuned quantitative agreement.

Third, the quasi-two-dimensional BKT description applies strictly in the anisotropic limit. Materials with strong interlayer coupling require a modified treatment, which may alter the detailed scaling of $T_c$.

### *8.4 Experimental Outlook*

The framework makes experimentally resolvable predictions.

Most prominently, the predicted $s^{\pm}$ symmetry in nickelates and the intermediate sensitivity to non-magnetic disorder provide near-term tests. Phase-sensitive ARPES, impurity-substitution studies, and measurements of magnetic excitation spectra under pressure offer independent probes of the underlying mechanism.

More broadly, systematic comparison of structure factor amplitudes, phase stiffness scaling, and gap symmetry across related material families can directly test the frustration-based selection principle.

### *8.5 Conceptual Implications*

If supported experimentally, the results suggest that superconductivity is not fundamentally defined by a specific pairing interaction, but by the existence of a stable composite class and the energetic cost of breaking global fiber coherence.

In this view, pairing is a geometric necessity in regimes where local frustration makes isolated excitations energetically unfavorable. Superconductivity then emerges as the collective phase-locking of frustration-stabilized composites.

Such an interpretation unifies mediator-driven and strongly correlated superconductors within a single structural framework, while remaining falsifiable through symmetry selection, stiffness scaling, and disorder response.

Future work should aim at deriving the fiber cost functional from explicit microscopic models, quantifying higher-order corrections to $r_{\mathcal{F}}$ scaling, and extending the framework to other unconventional superconductors, including iron-based compounds and nickelate variants.

## 9 Falsifiability and Failure Conditions

The present mechanism is structurally constrained and therefore falsifiable. It does not aim to accommodate all possible superconducting behaviors. It makes specific predictions whose failure would invalidate the core hypothesis.

**Nickelate gap symmetry.** If superconducting nickelates are conclusively shown to exhibit a robust $d_{x^2-y^2}$ ($B_{1g}$) gap symmetry with nodal structure identical to cuprates at optimal doping, while neutron or RIXS data confirm a substantially reduced $(\pi, \pi)$ frustration sector relative to cuprates, the real-space frustration selection criterion would be falsified.

**Absence of correlation between frustration and symmetry.** If materials with comparable frustration ratio $r_{\mathcal{F}}$ exhibit incompatible gap symmetries without additional orbital or lattice channels accounting for the discrepancy, the fiber-raccordement minimization hypothesis would be invalid.

**Impurity response outside the predicted window.** The framework predicts an intermediate suppression rate of $T_c$ under non-magnetic disorder for nickelates: stronger than $s^{++}$ immunity, weaker than Abrikosov–Gor'kov $d$-wave behavior. If systematic impurity-substitution studies yield suppression curves indistinguishable from either pure $d$-wave or pure $s^{++}$ limits, the proposed $s^{\pm}$ fiber mechanism would be ruled out.

**Breakdown of the $\rho_s \propto \delta J$ scaling.** In strongly correlated quasi-2D systems governed by projective frustration, the phase stiffness must scale as $\rho_s \propto \delta J$ to leading order. If high-resolution $\mu$SR and optical measurements demonstrate a scaling incompatible with this relation across the doping range, without fine-tuned cancellations, the mechanism would lose internal consistency.

**Pressure decoupling.** If pressure enhances $T_c$ in nickelates without measurable modification of either the magnetic exchange scale $J$ or the frustration proxy $S(\pi, \pi)$, the structural interpretation of pressure effects would fail.

In summary, the mechanism can be invalidated by experimental evidence disconnecting gap symmetry, frustration amplitude, phase stiffness, and disorder response. The predictive content of the framework lies precisely in this enforced coupling between real-space frustration geometry and superconducting order.

## 10 Conclusion

We have proposed a structural interpretation of superconductivity as projective phase locking of $w = 2$ composite configurations within a fiber-based geometric framework.

The mechanism does not rely on a specific pairing interaction. Instead, superconductivity emerges when composite formation reduces real-space frustration and global fiber coherence becomes energetically favorable.

In conventional superconductors, the projective gap is stabilized by mediator-assisted binding, recovering the London and Ginzburg–Landau structure. In strongly correlated systems, pairing arises from the reduction of staggered frustration, naturally separating amplitude and phase scales and accommodating pseudogap phenomena.

The symmetry of the superconducting order parameter is determined by a minimization principle on the fiber raccordement cost under lattice and frustration constraints. Applied to nickelates, this criterion predicts an extended $s^{\pm}$ symmetry, intermediate disorder sensitivity, and a critical temperature scaling set by independently measurable exchange and frustration scales.

The framework is explicitly falsifiable through gap symmetry determination, phase stiffness scaling, disorder response, and pressure dependence.

More broadly, the results suggest that superconductivity may be understood as a geometric constraint phenomenon: when isolated excitations become energetically frustrated, composite formation and collective phase locking provide the only admissible relaxation channel.

Further work is required to derive the cost functional from microscopic dynamics and to extend the analysis to additional material classes.

## A Operational Definition and Quantitative Grounding of the Frustration Proxy $\mathcal{F}$

### *A.1 Microscopic Definition*

The frustration density $\mathcal{F}$ measures the local inability of the spin background to simultaneously satisfy exchange bonds on a plaquette. On a square lattice with nearest-neighbour exchange $J_1 > 0$ and next-nearest-neighbour exchange $J_2 \geq 0$, the standard frustration parameter is

$$f_\square \;=\; \frac{J_2}{J_1}, \tag{39}$$

with $f_\square = 0$ corresponding to the unfrustrated Néel state and $f_\square = 1/2$ to the maximally frustrated point. In the correlated electron language relevant to cuprates and nickelates, the superexchange scales $J_i$ are related to the Hubbard parameters by $J_i = 4t_i^2/U$, so that $f_\square = (t_2/t_1)^2$.

At the macroscopic level we define the *frustration density* $\mathcal{F}$ as the spatial average of the local plaquette frustration weighted by the projective fiber cost:

$$\mathcal{F} \;\equiv\; \frac{1}{N_\square} \sum_\square f_\square \, \frac{C_\Pi(\mathbf{r}_\square)}{C_\Pi^{\max}}, \tag{40}$$

where $N_\square$ is the number of plaquettes and $C_\Pi(\mathbf{r}_\square)$ is the local raccordement cost evaluated at plaquette centre $\mathbf{r}_\square$.

### *A.2 Link to the Magnetic Structure Factor*

Because $\mathcal{F}$ as defined in Eq. (40) is not directly measurable, we identify its leading-order experimental proxy. The equal-time magnetic structure factor at wavevector $\mathbf{Q}$,

$$S(\mathbf{Q}) \;=\; \frac{1}{N}\sum_{i,j} e^{i\mathbf{Q}\cdot(\mathbf{r}_i-\mathbf{r}_j)}\langle \mathbf{S}_i\cdot\mathbf{S}_j\rangle, \tag{41}$$

is maximal at $\mathbf{Q}=(\pi,\pi)$ in the Néel state ($S_{\max}\simeq S(S+1)$) and is suppressed by frustration. The dominant frustration channel in the $CuO_2$ or $NiO_2$ plane is therefore encoded in $S(\pi,\pi)/S_{\max}$. To leading order in $f_\Box$, a self-consistent spin-wave calculation gives

$$\frac{S(\pi,\pi)}{S_{\max}} \;\approx\; 1-\alpha_0\, f_\Box + \mathcal{O}(f_\Box^2), \tag{42}$$

with $\alpha_0\sim 2$–$4$ depending on spin value and lattice geometry. The first-order proxy used throughout this work,

$$\mathcal{F} \;\propto\; \frac{S(\pi,\pi)}{S_{\max}}, \tag{43}$$

follows from combining Eqs. (42) and (40): to leading order, increasing $f_\Box$ decreases $S(\pi,\pi)/S_{\max}$ and decreases $\mathcal{F}$, consistently with the physical picture in which isotropization of frustration reduces the staggered channel. Higher-order corrections in $f_\Box$ renormalise $\alpha_0$ but do not alter the monotonic relation between $\mathcal{F}$ and $S(\pi,\pi)/S_{\max}$: the structure factor remains a strictly decreasing function of $f_\Box$ for all $f_\Box<1/2$, so the qualitative selection mechanism is unaffected unless fine-tuned cancellations occur (see Section 8.3).

### *A.3 Doping Dependence and Typical Values*

The carrier concentration $\delta$ modifies both the exchange scale and the magnetic order. To leading order in the $t$–$J$ model,

$$J_{\rm eff}(\delta) \;\approx\; J\,(1-2\delta), \qquad S(\pi,\pi;\delta) \;\approx\; S(\pi,\pi;0)\,(1-\beta\,\delta), \tag{44}$$

where $\beta\sim 3$–$5$ is extracted from neutron scattering and RIXS measurements. Combining Eqs. (43) and (44) gives the doping-dependent proxy

$$\mathcal{F}(\delta) \;\propto\; S(\pi,\pi;\delta) \;\approx\; \mathcal{F}_0\,(1-\beta\,\delta), \tag{45}$$

which decreases monotonically from $\mathcal{F}_0$ at half-filling toward zero in the heavily overdoped regime, consistent with the dome structure discussed in Section 5.5.

Representative values for the two material families considered in this work are collected in Table 1.

**Table 1.** Representative values of the frustration proxy $\mathcal{F}$ at optimal doping, extracted from neutron scattering and RIXS measurements of $S(\pi,\pi)$. $S_{\max}$ is normalised to the fully-ordered value at half-filling.

| Material | $J$ (meV) | $\delta_{\rm opt}$ | $S(\pi,\pi)/S_{\max}$ | $\mathcal{F}$ (proxy) |
|---|---|---|---|---|
| LSCO (optimal) | $\sim 130$ | 0.16 | 0.30–0.45 | large |
| YBCO (optimal) | $\sim 120$ | 0.16 | 0.25–0.40 | large |
| $La_3Ni_2O_7$ | $\sim 50$–$70$ | $\sim 0.15$ | 0.10–0.20 | moderate |

The cuprate values are consistent with strong staggered frustration, selecting $d_{x^2-y^2}$ symmetry. The nickelate values reflect a substantially reduced and partially isotropized frustration sector, consistent with the $s^\pm$ prediction of Section 6.

### *A.4 Relation between $\mathcal{F}$ and the Frustration Ratio $r_F$*

The dimensionless frustration ratio $r_F$ introduced in Eq. (31) of the main text is defined as the normalised frustration density:

$$r_F \;\equiv\; \frac{\mathcal{F}}{\mathcal{F}_{\rm crit}}, \tag{46}$$

where $\mathcal{F}_{\rm crit}$ is the value of $\mathcal{F}$ at the maximally frustrated point $f_\Box=1/2$ on the unfrustrated reference lattice. By construction $r_F\in[0,1]$, with $r_F=1$ corresponding to the Néel state at half-filling and $r_F=0$ to the fully isotropized or overdoped limit. Using Eq. (43), one obtains the experimental proxy

$$r_F \;\approx\; \frac{S(\pi,\pi)}{S(\pi,\pi)\big|_{\delta=0}}, \tag{47}$$

so that $r_F$ is directly accessible from the ratio of the measured structure factor at doping $\delta$ to its half-filling reference value. This identification makes the frustration ratio an observable quantity independent of any free fitting parameter.

## B Variational Argument for Symmetry Selection

### *B.1 Setup and Admissible Representations*

We formalize the symmetry selection principle of Section 6 through an explicit variational argument. Consider a quasi-two-dimensional system with point group $G = D_{4h}$ and a dominant frustration wavevector $\mathbf{Q} = (\pi, \pi)$. The gap function $\Delta(\mathbf{k})$ must belong to an irreducible representation (irrep) of $G$. The four leading irreps are

$$A_{1g}: \quad \Delta_s(\mathbf{k}) = \Delta_0, \; B_{1g}: \quad \Delta_d(\mathbf{k}) = \Delta_0(\cos k_x - \cos k_y), \; B_{2g}: \quad \Delta_{d'}(\mathbf{k}) = \Delta_0 \sin k_x \sin k_y, \; A_{1g}^{\pm}: \quad \Delta_{s^{\pm}}$$

### *B.2 The Raccordement Cost Functional*

The fiber raccordement cost functional of Eq. (30) takes the explicit form

$$C_{\rm tot}[\Delta] \;=\; \int_{\rm FS} \frac{d\Omega_{\mathbf{k}}}{(2\pi)^2} \Big[ r_F^2 \left|\Delta(\mathbf{k}+\mathbf{Q}) + \Delta(\mathbf{k})\right|^2 + (1 - r_F^2) \left|\nabla_{\mathbf{k}}\Delta(\mathbf{k})\right|^2 \Big], \tag{48}$$

where $r_F \in [0, 1]$ is the frustration ratio of Eq. (31) and the integral runs over the Fermi surface with solid-angle measure $d\Omega_{\mathbf{k}}$. The first term penalises configurations whose sign does not alternate under translation by $\mathbf{Q}$; the second penalises rapid phase variations along the Fermi surface.

### *B.3 Minimization: $d_{x^2-y^2}$ as the Global Minimum*

**Strongly staggered regime ($r_F \to 1$).** The dominant contribution to $C_{\rm tot}$ is the staggered mismatch term. For the $B_{1g}$ gap one verifies directly that

$$\Delta_d(\mathbf{k} + (\pi, \pi)) = \Delta_0\big(\cos(k_x + \pi) - \cos(k_y + \pi)\big) = -\Delta_0(\cos k_x - \cos k_y) = -\Delta_d(\mathbf{k}), \tag{49}$$

so that the staggered mismatch $|\Delta(\mathbf{k} + \mathbf{Q}) + \Delta(\mathbf{k})|^2$ vanishes identically for $B_{1g}$. By contrast, the $s$-wave and $s^{\pm}$ irreps both contribute $4r_F^2|\Delta_0|^2 > 0$. Therefore,

$$\operatorname*{arg\,min}_{\Delta\in{\rm irreps}(D_{4h})} \; C_{\rm tot}[\Delta] \;=\; B_{1g} \quad \text{for } r_F \gtrsim r_F^*. \tag{50}$$

**Weakly staggered regime ($r_F \to 0$).** The gradient term dominates. On the cuprate or nickelate Fermi surface, the extended $s^{\pm}$ form factor accumulates a smaller gradient cost than the pure $s$-wave because its nodes lie away from the principal Fermi arcs. For the nickelate multi-band geometry, inter-orbital hybridization further reduces $C_{\rm tot}$ in the $A_{1g}^{\pm}$ channel relative to $B_{1g}$, yielding

$$\operatorname*{arg\,min}_{\Delta} C_{\rm tot}[\Delta] \;=\; A_{1g}^{\pm} \quad \text{for } r_F \lesssim r_F^*. \tag{51}$$

### *B.4 Required Hypotheses*

The variational argument rests on three independently testable assumptions. First, the dominant frustration wavevector is $\mathbf{Q} = (\pi, \pi)$; incommensurate channels would shift the nodal structure. Second, the Fermi surface possesses $D_{4h}$ symmetry; orthorhombic distortions mix $B_{1g}$ and $B_{2g}$ and can deflect the nodes from the zone diagonal. Third, the cost functional Eq. (48) captures the leading symmetry-breaking term; higher-order corrections may renormalise $r_F^*$ but do not change the qualitative $B_{1g}$ versus $A_{1g}^{\pm}$ selection. All three assumptions are independently testable through neutron scattering, quantum oscillations, and gap-symmetry measurements.

## C Mapping from the Projective Gap $\Delta_\Pi$ to the ARPES Gap $\Delta(\mathbf{k})$

### *C.1 Definition Recalled*

The projective gap $\Delta_\Pi \equiv \lambda_{\min}(L_\Pi)|_{w=2}$ is the lowest positive eigenvalue of the linearised relaxation operator restricted to the $w = 2$ fiber sector (Eq. (7) of the main text). $L_\Pi$ acts on fiber-phase fluctuations $\delta\theta(\mathbf{r})$ around the coherent composite background; its spectrum encodes the local cost of fiber deformation.

*C.2 Isotropic Limit: Recovery of BCS*

In the absence of anisotropic frustration ($r_F = 0$), the relaxation operator $L_\Pi$ is spatially uniform, its eigenfunctions are plane waves $e^{i\mathbf{k}\cdot\mathbf{r}}$, and the eigenvalue is $\mathbf{k}$-independent at leading order:

$$\lambda(\mathbf{k})\big|_{r_F=0} \;=\; \Delta_\Pi^{(0)} \;=\; \text{const.} \tag{52}$$

This corresponds to a nodeless, fully isotropic quasiparticle gap, recovering the BCS identification $\Delta_{\rm BCS} = \Delta_\Pi^{(0)}$ of Section 3.2.

*C.3 Anisotropic Regime and Emergence of Nodal Structure*

When staggered frustration is present ($r_F > 0$), the operator acquires a $\mathbf{k}$-dependent correction:

$$L_\Pi(\mathbf{k}) \;=\; L_\Pi^{(0)} + r_F\, V(\mathbf{k};\mathbf{Q}), \tag{53}$$

where $V(\mathbf{k};\mathbf{Q})$ is the fiber-phase mismatch kernel evaluated at wavevector $\mathbf{Q}$. First-order perturbation theory gives

$$\lambda(\mathbf{k}) \;=\; \Delta_\Pi^{(0)} + r_F\, \langle \mathbf{k}\,|\, V(\cdot;\mathbf{Q})\,|\,\mathbf{k}\rangle + \mathcal{O}(r_F^2). \tag{54}$$

For $\mathbf{Q} = (\pi,\pi)$ and $B_{1g}$ symmetry, the matrix element evaluates to $\langle \mathbf{k}|V|\mathbf{k}\rangle \propto (\cos k_x - \cos k_y)$, so that

$$\Delta_\Pi(\mathbf{k}) \;\equiv\; \lambda(\mathbf{k}) \;\propto\; \Delta_\Pi^{(0)}\, |\cos k_x - \cos k_y|. \tag{55}$$

*C.4 Identification with the Measured ARPES Gap*

The quasiparticle gap measured by ARPES at Fermi momentum $\mathbf{k}_F$ equals the energy cost of creating a single $w = 1$ excitation from the composite, which equals $\lambda(\mathbf{k}_F)$. The identification is therefore

$$\Delta(\mathbf{k}) \;=\; \Delta_\Pi(\mathbf{k}) \;=\; \lambda(\mathbf{k})\big|_{w=2}. \tag{56}$$

Three consequences follow directly from Eq. (56).

**(1) Isotropic BCS case.** $\Delta(\mathbf{k}) = \Delta_\Pi^{(0)}$: nodeless and fully isotropic, consistent with Eq. (52).

**(2) Nodal *d*-wave (cuprates).** $\Delta(\mathbf{k}) \propto |\cos k_x - \cos k_y|$ vanishes along the zone diagonal ($k_x = \pm k_y$), reproducing the four nodal points observed in cuprate ARPES, with the maximal gap at $(\pi, 0)$ and $(0, \pi)$.

**(3) Extended $s^\pm$ (nickelates).** For $r_F < r_F^*$ the selected irrep is $A_{1g}^\pm$, giving $\Delta(\mathbf{k}) \propto |\cos k_x + \cos k_y|$: the gap changes sign between electron and hole pockets but is nodeless on each individual Fermi sheet, consistent with the intermediate disorder sensitivity predicted in Section 7.4.

The proportionality constant between $\Delta_\Pi^{(0)}$ and the observed maximum gap amplitude is set by the raccordement cost scale. From Section 4.1, $\Delta_\Pi^{\rm amplitude} \sim J$, consistent with the empirical ratio $\Delta_{\max}/k_BT_c \sim 4$–6 in cuprates when combined with the BKT relation $k_BT_c \simeq \frac{\pi}{2}\rho_s$ and the leading-order stiffness $\rho_s \sim \delta J$.

## D Quantitative Estimate of $T_c$ for Nickelates

*D.1 Calibration Strategy and the Ratio Formula*

Rather than introducing an independent proportionality constant with unclear physical content, we derive the nickelate $T_c$ directly from the ratio formula of Eq. (36) of the main text:

$$\frac{T_c^{\rm Ni}}{T_c^{\rm Cu}} \;\approx\; \frac{\delta^{\rm Ni}\, J^{\rm Ni}}{\delta^{\rm Cu}\, J^{\rm Cu}} \cdot \frac{r_F^{\rm Ni}}{r_F^{\rm Cu}}, \tag{57}$$

where the linear approximation $f(r_F)/f(r_F^{\rm Cu}) \approx r_F/r_F^{\rm Cu}$ holds at leading order in $r_F$ (see below for the sensitivity to this assumption). We take LSCO at optimal doping as the calibration reference: $T_c^{\rm Cu} = 38$ K ($k_BT_c^{\rm Cu} = 3.27$ meV), $J^{\rm Cu} = 130$ meV, $\delta_{\rm opt}^{\rm Cu} = 0.16$, $r_F^{\rm Cu} = 0.80$. No free parameter is adjusted; all quantities on the right-hand side of Eq. (**??**) are independently measurable.

*D.2 Sensitivity to the Linear Approximation for $f(r_F)$*

The approximation $f(r_F) \propto r_F$ is the simplest monotonic ansatz consistent with the boundary conditions $f(0) = 0$ and $f(1) = 1$. Sub-linear behaviour $f(r_F) \sim r_F^\alpha$ with $\alpha < 1$ would increase the predicted $T_c^{\rm Ni}$; super-linear behaviour with $\alpha > 1$ would decrease it. Given the present absence of a fully microscopic derivation of $f$, we treat $r_F^{\rm Ni}$ as an effective parameter that absorbs the uncertainty in $f$: the range $r_F^{\rm Ni} \in [0.20, 0.35]$ used below encompasses both the direct RIXS estimate of $S^{\rm Ni}(\pi,\pi)/S^{\rm Cu}(\pi,\pi)$ and the leading sub-linear correction. This range is conservative and bounded from above by measurements of the $(\pi,\pi)$ spectral weight in infinite-layer nickelates.

### *D.3 Explicit Calculation for Ambient-Pressure Infinite-Layer Nickelates*

For $R_{1-x}Sr_xNiO_2$ at optimal doping, the experimentally constrained input parameters are

$$J^{\rm Ni} \simeq 60\text{–}80 \text{ meV } (696\text{–}928 \text{ K}), \qquad \delta^{\rm Ni}_{\rm opt} \simeq 0.20, \qquad r^{\rm Ni}_F \simeq 0.20\text{–}0.35, \tag{58}$$

where the meV-to-Kelvin conversions use 1 meV = 11.60 K. Inserting the central values $J^{\rm Ni} = 70$ meV and $r^{\rm Ni}_F = 0.25$ into Eq. (**??**) gives the explicit numerical result

$$T^{\rm Ni}_c = 38 \text{ K} \times \frac{0.20 \times 70}{0.16 \times 130} \times \frac{0.25}{0.80} = 38 \text{ K} \times 0.673 \times 0.313 \simeq 8.0 \text{ K}, \tag{59}$$

in good agreement with the observed $T_c \simeq 9$–15 K. Propagating the full input ranges of Eq. (**??**) yields

$$T^{\rm Ni}_c \simeq 5\text{–}12 \text{ K}, \tag{60}$$

where the lower bound corresponds to ($J = 60$ meV, $r_F = 0.20$) and the upper bound to ($J = 80$ meV, $r_F = 0.35$). The overlap with the experimental range is summarised in Table **??**.

**Table 2.** Predicted and observed $T_c$ for nickelate families. All input parameters $(J, \delta_{\rm opt}, r_F)$ are taken from RIXS, neutron scattering, and transport measurements; see Eq. (**??**). No parameter is adjusted after calibration on LSCO. Conversion: 1 meV = 11.60 K.

| Material | $J$ (meV) | $\delta_{\rm opt}$ | $r_F$ | $T_c^{\rm pred}$ (K) | $T_c^{\rm exp}$ (K) |
|---|---|---|---|---|---|
| LSCO (calibration) | 130 | 0.16 | 0.80 | 38 | 38 |
| $R_{1-x}Sr_xNiO_2$ | 60–80 | 0.20 | 0.20–0.35 | 5–12 | 9–15 |
| $La_3Ni_2O_7$ (amb.) | 50–70 | 0.15 | 0.15–0.25 | 3–7 | $\sim 6$–10 |

The predicted range slightly underestimates the experimental upper end (15 K) for $r^{\rm Ni}_F$ at the low end of the RIXS-constrained window. This discrepancy is within the expected accuracy of the linear approximation for $f(r_F)$ and of the single-band treatment that neglects interlayer coupling in the bilayer geometry.

### *D.4 Pressure Enhancement: a Conditional Prediction*

Under applied pressure $P \gtrsim 14$ GPa, $La_3Ni_2O_7$ exhibits $T_c \simeq 80$ K in bulk crystals. Within the present framework, this enhancement requires a simultaneous increase of both $J(P)$ and $r_F(P)$. Reproducing $T_c(P) \simeq 80$ K from Eq. (**??**) with $\delta^{\rm Ni} = 0.25$ (bilayer effective doping) requires

$$J^{\rm Ni}(P)\, r^{\rm Ni}_F(P) \gtrsim 140 \text{ meV}, \tag{61}$$

for example $J(P) \simeq 200$ meV with $r_F(P) \simeq 0.70$, or $J(P) \simeq 180$ meV with $r_F(P) \simeq 0.80$. These parameter values are at the high end of what pressure is expected to induce based on *ab initio* estimates of orbital overlap, and the prediction is therefore conditional on future RIXS measurements of $J(P)$ and $S(\pi, \pi; P)$ under pressure. We emphasise that Eq. (**??**) provides the experimentally testable constraint: if pressure-dependent RIXS yields the product $J(P)\, r_F(P)$ significantly below 140 meV, the frustration mechanism would require a supplementary contribution to account for the observed $T_c(P)$.

### *D.5 Falsifiability of the Numerical Estimate*

The quantitative predictions above would be invalidated by the following observations. If $T^{\rm Ni}_c$ exceeds 20 K at ambient pressure without a corresponding increase of $J^{\rm Ni}$ or $\delta$, the $\rho_s \propto \delta J$ scaling would be broken. If pressure-dependent RIXS confirms $J(P) < 160$ meV and $r_F(P) < 0.60$ while $T_c(P) \simeq 80$ K is maintained, Eq. (**??**) would be violated and a mechanism beyond frustration renormalization would be required. If the pressure-induced $T_c$ enhancement is discontinuous without a corresponding jump in $S(\pi, \pi; P)$, the smooth frustration-driven picture of Eq. (**??**) would be insufficient. These constitute near-term falsifiable tests accessible with existing RIXS and transport techniques.

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